Week 5

Newtons first Law

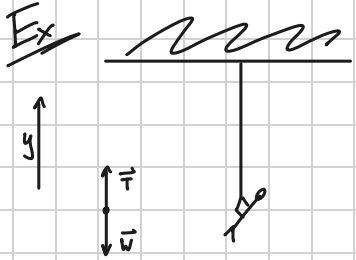
⇒ Particle in equilibrium

Condition for equilibrium $\sum \vec{F} = 0$

in 2D $x: \sum F_x = 0$

$y: \sum F_y = 0$

Ex



Assume rope has zero mass

Equilibrium.

$\vec{T} + \vec{w} = 0$

$y: \quad w = -mg$

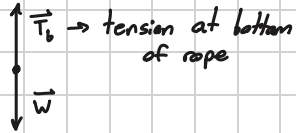
$T - mg = 0 \Rightarrow \boxed{T = mg}$

Ex

rope has mass m_r

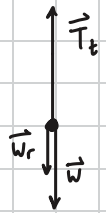


FBD for athlete alone



$T_b - mg = 0 \Rightarrow T_b = mg$

FBD for athlete + rope system



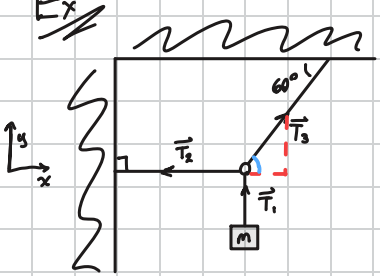
$\vec{T}_t$ – tension at top of rope

$\vec{T}_t + \vec{w} + \vec{w}_r = 0$

$w = -mg$, $w_r = -m_r g$

$\Rightarrow T_t - mg - m_r g = 0 \Rightarrow$ $\boxed{T_t = (m + m_r)g}$

Ex



the string and ring are massless. find tension.

Forces on ring: $\vec{T}_1$, $\vec{T}_2$, $\vec{T}_3$

Ring is in equilibrium: $\vec{T}_1 + \vec{T}_2 + \vec{T}_3 = 0$

x: $-T_2 + T_3 \cos\theta = 0$ (1)

y: $-T_1 + T_3 \sin\theta = 0$ (2)

FBD for mass m:



$T_1 = mg$

Sub into (2): $-mg + T_3 \sin\theta = 0$

$\Rightarrow T_3 = \frac{mg}{\sin\theta}$

Now Sub into ①

$$-T_2 + \frac{mg}{\sin\theta} \cdot \cos\theta = 0$$

$$\Rightarrow \quad T_2 = \frac{mg\cos\theta}{\sin\theta} = mg\cot\theta$$

Newtons Second Law

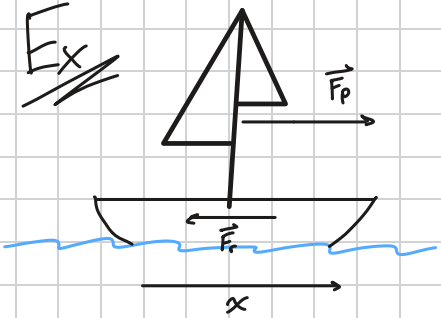
$$\sum \vec{F} = m\vec{a}$$

2D problem $\Rightarrow$

$$X: \sum F_x = ma_x$$

$$Y: \sum F_y = ma_y$$

Ex



$m = 200kg$

$F_r = 100N$

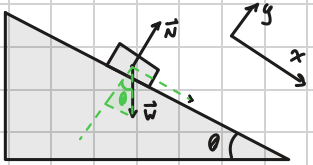
$a_x = 1.5 m/s^2$

What is F_p?

$X: \quad F_p - F_r = ma_x \Rightarrow F_p = F_r + ma_x \Rightarrow F_p = 400N$

Ex

No friction



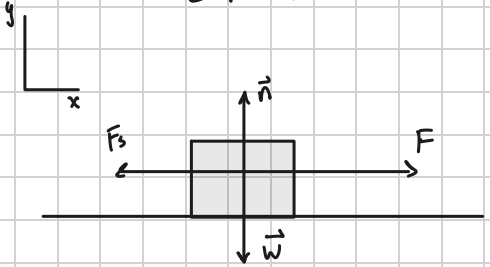
$\vec{w} + \vec{N} = m\vec{a}$

$y: \quad -mg\cos\theta + N = 0$

$x: \quad mg\sin\theta = ma$

$\Rightarrow \boxed{a = g\sin\theta}$

Static vs kinetic friction



no motion

$y: \vec{n} + \vec{w} = 0$

$n - mg = 0$

$\Rightarrow n = mg$

$X: \vec{F} + \vec{F}_s = 0$

$x: F - F_s = 0 \quad \Rightarrow F_s = F$

there a maximum friction force

$F_{smax} = \mu_s n \quad \Rightarrow F_s \leq \mu_s n$

if $F > F_{smax} \Rightarrow$ Object starts moving

in practice $\mu_k < \mu_s$

$F_k = \mu_k n$

Example A crate weighs 500N

A force of 200N is needed for const vel

with initial of 230N needed

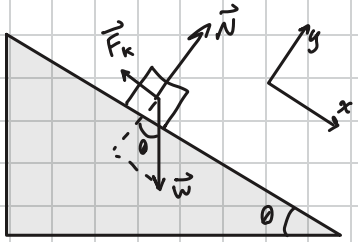
$$F_{s\,max} = \mu_s \cdot mg = 230N$$

$$\mu_s = \frac{230N}{500N} = 0.460$$

Once moving: $F_K = \mu_K \cdot mg = 200$

$$\Rightarrow \mu_k = \frac{200N}{500N} = 0.400$$

Ex



$w_x = w\sin\theta$

$w_y = w\cos\theta$

$w_x = mg\sin\theta$

$w_y = -mg\cos\theta$

Newtons second Law:

$\vec{w} + \vec{n} + \vec{F}_K = m\vec{a}$

x: $mg\sin\theta - \mu_K n = ma$

y: $n - mg\cos\theta = 0 \quad \Rightarrow \quad n = mg\cos\theta$

$\Rightarrow mg\sin\theta - \mu_K mg\cos\theta = ma$

$mg(\sin\theta - \mu_K\cos\theta) = ma$

$\Rightarrow \boxed{a = g(\sin\theta - \mu_K\cos\theta)}$

Note: formula only works

if $a > 0$ $(\sin\theta > \mu_K\cos\theta)$

or $\mu_K < \frac{\sin\theta}{\cos\theta} = \tan\theta$

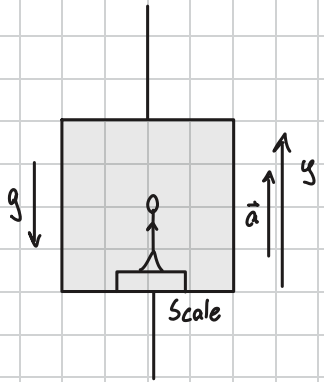
if $a = 0$

$$\boxed{a = g(\sin\theta - \mu_k \cos\theta)} = 0$$

$$\Rightarrow \quad \mu_k = \tan\theta$$

Ex

Apparent weight in an elevator





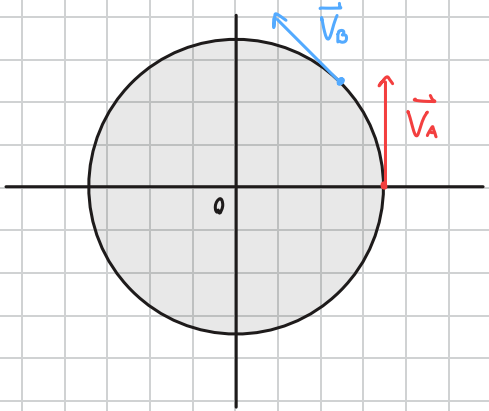
$$\vec{n} + \vec{w} = m\vec{a}$$

$$n - mg = ma$$

$$\Rightarrow \quad n = mg + ma = m(g + a)$$

$$a > 0 \Rightarrow n > mg$$

Note: the scale is measuring the normal force



$V = |\vec{V}| = \text{Constant}$

$\vec{a} = \vec{a}_t + \vec{a}_n$

$\vec{a} = \vec{a}_n = \vec{a}_{rad} = \vec{a}_{cp}$

radial or centripetal acceleration

$$\vec{F}_{cp} = m\vec{a}_{cp}$$

Centripetal force

$$a_{cp} = \frac{V^2}{R}$$

$$T = \frac{2\pi R}{V} \Rightarrow V = \frac{2\pi R}{T}$$

$$a_{cp} = \frac{W^2R^2}{R} = W^2R$$

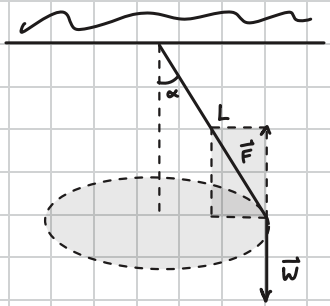
$$W = \frac{2\pi}{T} \quad \text{(angular velocity)}$$

or

$$V = WR \qquad W = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t}$$

$$a_{cp} = \frac{4\pi^2R^2}{T^2} \cdot \frac{1}{R} = \frac{4\pi^2R}{T^2}$$

Example





$R = L\sin\alpha$

$\vec{W} + \vec{F} = m\vec{a}$

$y: \; -mg + F\cos\alpha = 0 \Rightarrow \boxed{F = \frac{mg}{\cos\alpha}}$

$X: \; F\sin\alpha = \frac{mv^2}{R} = \frac{mv^2}{L\sin\alpha}$

Sub the F into the x equation:

$\frac{mg}{\cos\alpha}\sin\alpha = \frac{mv^2}{L\sin\alpha} \Rightarrow$ Solve for v^2:

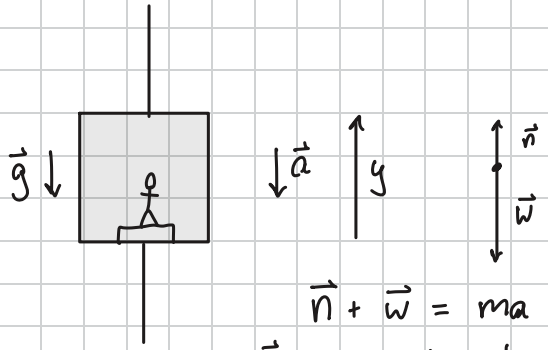
$v^2 = \frac{gL\sin^2\alpha}{\cos\alpha} \Rightarrow v = \sqrt{\frac{gL}{\cos\alpha}} \cdot \sin\alpha$

But we know that $T = \frac{2\pi L\sin\alpha}{v}$

$\Rightarrow T = \frac{2\pi L\sin\alpha}{\sqrt{\frac{gL}{\cos\alpha}} \cdot \sin\alpha} \Rightarrow T = 2\pi L\sqrt{\frac{\cos\alpha}{gL}}$

$\Rightarrow \boxed{T = 2\pi\sqrt{\frac{L\cos\alpha}{g}}}$

$a_{cp} = \frac{v^2}{R}, \quad F_{cp} = \frac{mv^2}{R}$



$\vec{n} + \vec{w} = m\vec{a}$

$\vec{a}$ is pointing downwards

$n - mg = -ma$ $\quad a > 0$

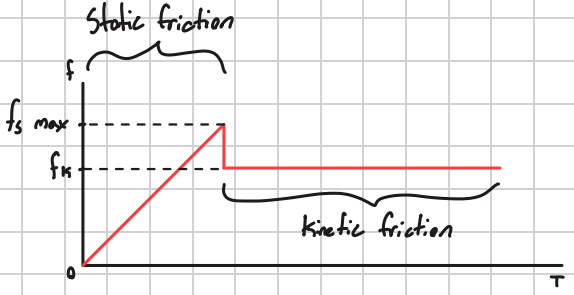
$\Rightarrow n = mg - ma = m(g-a)$

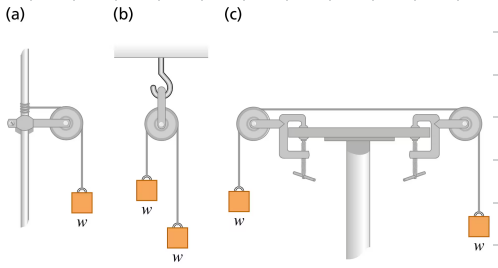
$\Rightarrow n = m(g-a) < mg$

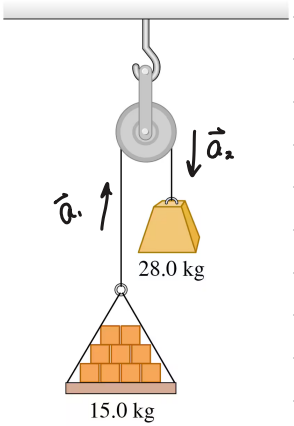
Read 5.5 from textbook

$f_k = \mu_k n$

$f_s \leq f_{s,max} = \mu_s n$



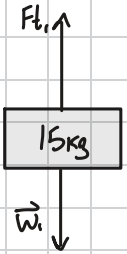




Draw two free-body diagrams, one for the load of bricks and one for the counterweight.

What is the magnitude of the upward acceleration of the load of bricks?

What is the tension in the rope while the load is moving? How does the tension compare to the weight of the load of bricks? To the weight of the counterweight?





$\vec{F}_{t_1} = \vec{w}_2$

$\vec{F}_{t_2} = \vec{w}_1$

find a_1

$F_{net} = ma$

$\vec{w}_2 = (28)(9.8) = 274.4N$

$\vec{w}_1 = (15)(9.8) = 147N$

$F_{net_{y_1}} = 274.4 - 147 = 127.4N$

moving with both masses so we have to add

$F_{net} = (m_1 + m_2) \cdot a$

$127.4 = (15+28) \cdot a \Rightarrow a = \frac{127.4}{(28+15)}$

$= 2.96\ m/s^2$